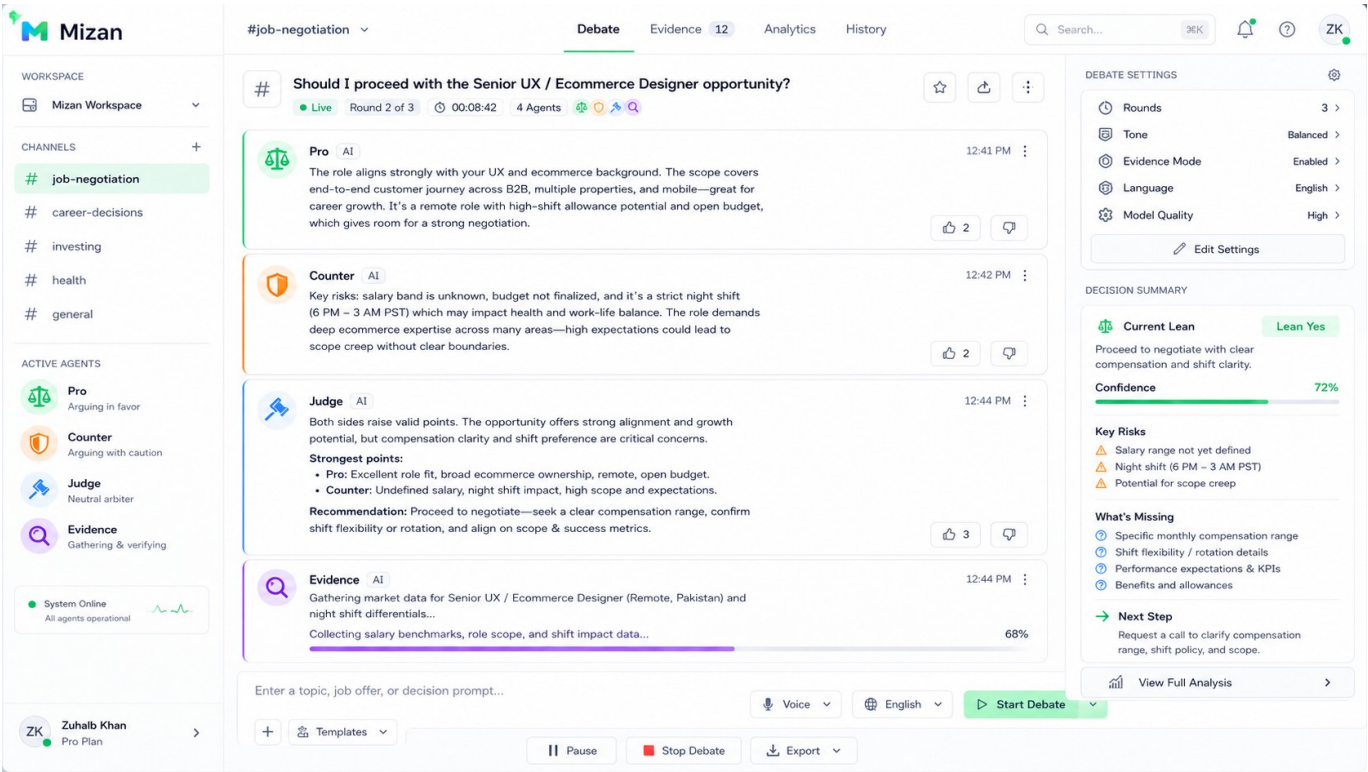


MIZAN

DECISION ROOM

Product Requirements, UX Specification, Agent Protocol and
Technical Blueprint

Authoritative implementation document for Codex



Product owner	Zak
Working brand	Mizan - Decision Room
Initial expert pack	Career and Hiring
Document version / date	1.0 / 25 July 2026

Two perspectives. One clearer decision.

Document Authority and Build Instruction

Codex directive

Treat this PDF as the primary product and implementation specification. Build the MVP end-to-end, preserve the architecture for future expert packs, and do not hard-code the product as an HR-only application. Where a requirement is marked "MVP", it is mandatory. Where it is marked "Later", create an extension point but do not overbuild it.

The product is a real-time decision-support environment where multiple AI agents examine one user-provided topic from opposing positions, verify evidence, identify missing information, and produce a practical decision. The interface borrows the channel-based readability of classic IRC clients while using a modern, light, production-grade SaaS visual system.

The first commercial use case is career and hiring. The underlying engine must remain domain-agnostic so that finance, health, business, relationships, legal, technology and general-life expert packs can be added later without changing the debate runtime or the primary interface shell.

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1. Product Vision and Positioning

1.1 Product definition

Mizan - Decision Room is a structured multi-agent deliberation platform. A user opens a channel, provides a decision topic and relevant context, and watches bounded AI agents construct a thesis, an antithesis, evidence checks and a final adjudication. The output is not merely a summary; it is a decision record containing the strongest arguments, verified facts, assumptions, unresolved unknowns, confidence and a concrete next step.

1.2 Brand architecture

Element	Decision
Primary brand	Mizan
English descriptor	Decision Room
Tagline	Two perspectives. One clearer decision.
Initial workspace	Career and Hiring
Initial channel example	#job-negotiation
Default theme	Light
Optional theme	Dark

Naming rule

Use "Mizan" in the interface. Use "Decision Room" as the English explanatory descriptor. Do not rename the entire product to an HR-specific title.

1.3 Strategic differentiation

- The product does not simply ask two models to disagree. It controls disagreement through roles, shared evidence, round limits and a judge contract.
- It detects whether domain expertise exists before allowing authoritative language.
- It separates facts, user-provided claims, model assumptions, inferences and unknowns.
- It produces a practical next action rather than an endless balanced essay.
- It preserves a full decision trail that the user can revisit, export and compare.

2. Problem Definition and Product Principles

2.1 User problem

Today a user who wants opposing AI analysis must manually copy one model's answer into another model, then copy the response back again. This wastes time, destroys conversational continuity, creates inconsistent context and encourages the models to critique each other instead of solving the user's decision.

2.2 Product outcome

The product must run the entire dialectical process in one live channel, expose what each agent is doing, allow the user to intervene, and conclude within a predictable time and cost budget.

2.3 Non-negotiable principles

Principle	Implementation consequence
No fake expertise	Show domain readiness and switch to clarification, research, general reasoning or qualified-expert-required mode.
Bounded debate	Every session has a maximum round count, token budget, time budget and cost budget.
Evidence before authority	High-stakes or current factual claims require evidence or must be labelled unverified.
User stays in control	The user can pause, stop, interject, change criteria and request an early verdict.
Decision, not theatre	The judge must choose, conditionally choose, or state that evidence is insufficient.
Provider independence	No business logic may depend on one model vendor.
Language integrity	A selected language mode applies to the entire session; do not mix languages inside messages unless explicitly requested.
Traceability	Every verdict must link back to arguments, evidence items and assumptions.

3. Users, Jobs and Decision Contexts

3.1 Primary user for MVP

A professional who receives job opportunities, recruiter messages, interview requests or offers and wants immediate strategic analysis, exact replies, interview preparation and negotiation guidance.

3.2 Secondary future users

User type	Job to be done	Required mode
General consumer	Compare difficult life choices without manually consulting multiple models.	General reasoning
Founder / manager	Examine product, hiring or business decisions from opposing commercial perspectives.	Business expert pack
Investor	Evaluate an investment thesis against risks and current evidence.	Finance expert pack + live data
Patient / caregiver	Structure questions and options without replacing a clinician.	Health pack + qualified-expert boundary
Designer / engineer	Stress-test product and technical decisions.	Product / technology expert pack

3.3 Core jobs to be done

- When I have a complex decision, help me see the strongest case for and against it without losing context.
- When claims conflict, show me which claims are supported, assumed or unknown.
- When important information is missing, ask only the questions that can materially change the decision.
- When the debate ends, give me a decisive, confidence-calibrated next action.
- When I return later, let me reopen the complete channel and continue from the existing decision record.

4. Scope: MVP, Version 1 and Later Roadmap

4.1 MVP - mandatory

- Email/password sign-in or a secure single-user mode configurable by environment.
- Workspace and channel-based interface.
- Create a topic from text; attach contextual text or files as references.
- Language modes: English, Urdu and Roman Urdu.
- Four logical agents: Thesis, Antithesis, Evidence and Judge.
- Career and Hiring Expert Pack.
- Domain router, risk rating and expert-pack availability status.
- Clarification mode before debate when decision-critical information is missing.
- Configurable debate rounds: 1 to 6; default 3.
- Real-time streamed messages with pause, stop and user interjection.
- Decision Summary panel with current lean, confidence, key risks, missing information and next step.
- History, search and export to Markdown, JSON and printable PDF.
- Light theme as default and dark theme as optional.
- Provider adapters for at least two different LLM providers or two separately configured models.
- Basic usage, latency, token and cost telemetry.

4.2 Version 1 - build-ready extension points

- Voice input and text-to-speech with separate agent voices.
- Web research connectors and URL evidence ingestion.
- Expert Pack management screen.
- Saved debate templates.
- Weighted decision criteria.
- Decision comparison across multiple sessions.
- Team workspaces and shareable read-only debate links.
- Optional human expert handoff.

4.3 Explicitly out of scope for MVP

- Autonomous medical, legal or financial instructions.
- Public marketplace of experts.
- Native iOS or Android applications.
- Unlimited agent swarms.
- Autonomous emailing, messaging or job application submission.
- Complex enterprise billing and organisation administration.
- Fully automated web research without user-visible evidence controls.

5. Functional Architecture

Mizan - System Architecture

Provider-agnostic, real-time, evidence-aware decision orchestration



5.1 Core services

Service	Responsibility	MVP
Topic Router	Classify domain, decision type, risk and expert-pack availability.	Yes
Clarifier	Identify missing information that could change the verdict.	Yes
Expert Pack Loader	Load domain prompts, evaluation criteria, safety policy and tool access.	Yes
Debate Orchestrator	Control rounds, turns, context windows, retries, pause/stop and user interjection.	Yes
Evidence Service	Store references, extract claims and assign verification status.	Yes - supplied evidence
Research Connector	Search external sources and build cited evidence bundles.	Later
Judge Service	Produce adjudication using structured schema.	Yes
Export Service	Render Markdown, JSON and PDF decision records.	Yes
Voice Gateway	Speech-to-text and text-to-speech.	Later

5.2 Source-of-truth hierarchy

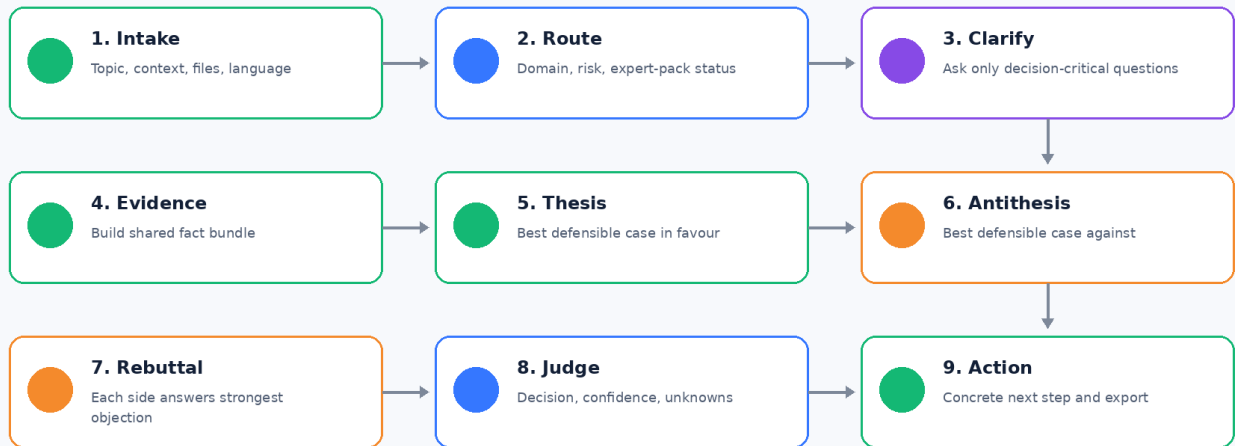
22. User-provided documents and explicit facts for the session.

23. Verified evidence items from connected research tools.
24. Expert Pack rules and evaluation framework.
25. Model reasoning and inference, always labelled as inference.
26. Unsupported model memory, which must never be presented as verified fact.

6. Runtime Debate Protocol

Runtime Debate Protocol

A bounded sequence designed to prevent endless AI argument loops



6.1 State machine

```

DRAFT -> ROUTING -> CLARIFICATION_REQUIRED | READY
READY -> EVIDENCE_PREP -> DEBATING
DEBATING -> PAUSED | STOPPED | JUDGING
PAUSED -> DEBATING | STOPPED
JUDGING -> COMPLETED | INSUFFICIENT_EVIDENCE | EXPERT_REQUIRED
COMPLETED -> REOPENED (new round with preserved history)
  
```

6.2 Default three-round sequence

Round	Thesis	Antithesis	Evidence	Judge
1 - Opening	Strongest case in favour.	Strongest case against.	Flag claims needing support.	Summarise dispute and identify missing facts.
2 - Rebuttal	Answer the strongest counterargument.	Answer the strongest pro argument.	Verify or label contested claims.	Score argument quality and remaining uncertainty.
3 - Closing	Final conditional recommendation.	Final failure case and safeguards.	Publish evidence status.	Return final decision contract.

6.3 Loop prevention

- Agents may not repeat an argument unless they add new evidence or a materially different inference.

- Every message must include argument IDs and the claims it addresses.
- The orchestrator rejects messages whose semantic similarity to the same agent's previous turn exceeds the configured threshold and requests a concise revision.
- The judge may end the debate early when no new decision-relevant information appears.
- A hard cap applies to rounds, tokens, elapsed time and provider cost.

6.4 User intervention

The user may post a message inside the live channel. The orchestrator classifies it as additional evidence, a changed constraint, a direct question, a correction or a command. If it materially changes the decision, the current round is marked superseded and agents receive the update before continuing.

7. Agent Roles and Prompt Contracts

7.1 Thesis agent

Construct the strongest defensible case in favour of the proposed action. It must not flatter the user, hide risks or fabricate evidence. It should state conditions under which its own recommendation fails.

7.2 Antithesis agent

Construct the strongest defensible case against the proposed action. It must identify hidden costs, weak assumptions, downside scenarios, reversibility and safer alternatives. It may not manufacture opposition merely to appear balanced.

7.3 Evidence agent

Build and maintain the shared claim ledger. For each claim, store its source, status, freshness, relevance and any contradiction. In MVP it works on supplied files and user context. External web research is an extension point.

7.4 Judge

Evaluate argument quality, evidence quality, user constraints, risk, reversibility and missing information. The judge must not average the agents. It must issue one of four verdict types: proceed, do not proceed, proceed conditionally, or insufficient information / qualified expert required.

7.5 Structured agent response contract

```
{
  "messageType": "opening|rebuttal|closing|evidence|judgment",
  "summary": "human-readable message",
  "claims": [
    {
      "claimId": "clm...",
      "text": "...",
      "status": "verified|user_provided|inference|assumption|unknown",
      "evidenceIds": ["ev..."],
      "confidence": 0.0
    }
  ],
  "addressesArgumentIds": ["arg..."],
  "newInformation": true,
  "decisionImpact": "high|medium|low"
}
```


7.6 Judge output contract

```
{
  "verdict": "proceed|do_not_proceed|proceed_conditionally|insufficient_information|expert_required",
  "headline": "...",
  "confidence": 0.72,
  "currentLean": "lean_yes|neutral|lean_no",
  "strongestProArguments": ["arg_..."],
  "strongestCounterArguments": ["arg_..."],
  "verifiedFacts": ["clm_..."],
  "assumptions": ["clm_..."],
  "keyRisks": ["..."],
  "missingInformation": ["..."],
  "conditions": ["..."],
  "nextAction": "...",
  "explanation": "..."
}
```

8. Expert Pack System and Domain Readiness

8.1 Purpose

An Expert Pack is a versioned configuration bundle that converts general model reasoning into domain-disciplined analysis. It is not a fine-tuned model. It contains prompts, decision criteria, required questions, safety constraints, tool permissions, evidence standards, terminology and output templates.

8.2 Expert Pack manifest

```
{
  "id": "career-hiring",
  "version": "1.0.0",
  "domains": ["career", "recruiting", "job_offer", "interview"],
  "riskPolicy": "moderate",
  "requiredContextFields": ["role", "company", "employmentType", "workHours"],
  "decisionCriteria": ["role_fit", "compensation", "schedule", "risk", "growth"],
  "tools": ["files", "web_optional"],
  "promptFiles": {
    "thesis": "thesis.md",
    "antithesis": "antithesis.md",
    "judge": "judge.md",
    "clarifier": "clarifier.md"
  }
}
```

8.3 Domain readiness states

State	Meaning	Allowed behaviour
Expert Pack Available	A domain-specific pack exists and required context is sufficient.	Run expert-guided debate.
Research Required	The decision depends on current external facts.	Pause authoritative debate until evidence is gathered or user accepts limited mode.
General Reasoning Only	The topic is subjective or no pack exists.	Structure trade-offs and label output as reasoning, not expert determination.
Qualified Expert Required	High-stakes or regulated decision	Provide questions, options and referral

State	Meaning	Allowed behaviour
	exceeds safe AI authority.	guidance; do not issue an authoritative verdict.
Insufficient Context	Key facts can materially alter the result.	Ask concise clarification questions before debate.

8.4 High-stakes rule

Safety boundary

For medical, legal, financial, crisis or other high-stakes topics, the router may allow structured discussion but must block language that claims professional diagnosis, legal determination, guaranteed returns or emergency authority.

9. Career and Hiring Expert Pack

9.1 Supported intents

- Analyse a recruiter or HR message.
- Decide whether to apply, reply, take a call, continue interviews or accept an offer.
- Assess role fit, seniority, title, employer and recruiter credibility.
- Analyse compensation structure, salary anchoring, currency, benefits and contract terms.
- Analyse working hours, remote arrangement and lifestyle cost.
- Generate an exact reply in the user's selected language and tone.
- Prepare interview answers and portfolio positioning.
- Compare multiple offers and define a walk-away point.

9.2 Required decision dimensions

Dimension	Questions
Role fit	Does the scope match the user's experience, level and target direction?
Employer quality	Is it a direct employer, agency, referral funnel or contract intermediary?
Compensation	Known band, currency, gross/net, benefits, current-salary anchoring and market basis.
Schedule	Timezone, fixed or flexible hours, shift cost and live-collaboration requirement.
Employment structure	Permanent, contract, outsourced, probation, notice and termination terms.
Growth and portfolio	Ownership, strategic influence, learning and demonstrable outcomes.
Risk and opportunity cost	Time spent, unpaid tests, career lock-in and alternatives.
Negotiation leverage	Information asymmetry, timing, competing candidates and disclosure strategy.

9.3 HR-specific output

- Proceed / proceed conditionally / reject recommendation.
- Role-fit score with evidence and gaps.
- Red flags separated from normal recruitment behaviour.
- Exact next message that sounds human, not over-engineered.
- What not to disclose yet.
- Questions to ask on the next call.
- Expected interview stages and preparation plan.
- Negotiation target, acceptable zone and walk-away conditions only when evidence supports them.

9.4 Critical honesty rule

The pack must never invent company salary bands, market rates, recruiter motives or employer intentions. It must label each statement as verified fact, recruiter claim, reasonable estimate, inference or unknown.

10. Information Architecture and Navigation

10.1 Primary shell

Region	Content
Top bar	Current channel, Debate, Evidence, Analytics, History, search, notifications, help, account.
Left rail	Workspace, channels, active agents, system status, user/account.
Main channel	Topic header, live status, round/timer, agent messages, user interventions, composer.
Right rail	Debate settings, decision summary, confidence, risks, missing information and next step.
Bottom controls	Pause, stop, export, voice, language and start debate.

10.2 Channel model

Each decision topic is a channel. Channels belong to a workspace and contain a topic, context, attachments, configuration, messages, evidence items, decisions and follow-up rounds. The default workspace may contain channels such as #job-negotiation, #career-decisions, #investing, #health and #general.

10.3 Slash commands

Command	Behaviour
/topic	Create or edit the current decision topic.
/rounds 4	Set the maximum debate rounds.
/pause	Pause after the current streamed event.
/resume	Resume the current session.
/stop	Stop all pending agent jobs.

Command	Behaviour
/evidence	Open evidence mode or request verification.
/verdict	End debate and ask the judge for the best current verdict.
/language urdu	Change the session language after confirmation.
/export pdf	Generate a decision record.
/criteria	Edit weighted decision criteria.

11. Screen-by-Screen UX Specification

11.1 Authentication

Minimal sign-in screen with Mizan branding, email, password, forgot password and optional single-user bypass controlled by environment configuration. Do not add social-login complexity in MVP unless it is trivial within the chosen authentication library.

11.2 Onboarding

- Choose preferred language: English, Urdu or Roman Urdu.
- Choose default decision style: concise, balanced or deep.
- Choose default model providers when more than one is configured.
- Create first workspace and optionally load the Career and Hiring starter template.
- Explain that Mizan distinguishes evidence, assumptions and uncertainty.

11.3 Workspace home

- Recent decision channels with status, verdict, confidence and last activity.
- Create Decision button.
- Quick templates: recruiter message, job offer, interview preparation, general decision.
- Usage summary and system/provider status.
- No decorative analytics that do not help the user take action.

11.4 Create decision modal

Field	Specification
Topic	Required, plain-language decision question.
Context	Optional rich text or pasted conversation.
Attachments	PDF, DOCX, TXT, image and supported text extraction.
Expert Pack	Auto-detected with manual override.
Language	English, Urdu, Roman Urdu.
Rounds	1-6, default 3.
Tone	Concise, balanced, adversarial, executive.
Evidence mode	Supplied evidence only in MVP; external research later.
Decision criteria	Optional weighted criteria; defaults from Expert Pack.

Field	Specification
Budget controls	Maximum time, tokens and estimated provider cost.

11.5 Live debate channel

This is the core screen represented in the approved light-theme reference. Agent messages must be readable as a continuous channel, not as isolated modal dialogs. Each message displays agent identity, role, timestamp, round, claim status, evidence links and reactions. Streaming must remain stable when the right rail is open.

- Topic header with status, round, elapsed time and active agents.
- Agent message cards with controlled accent colour: Thesis green, Antithesis orange, Judge blue, Evidence purple.
- User messages visually distinct but within the same chronological feed.
- Composer supports text, attachments, slash commands and future voice input.
- Pause, stop and early verdict remain visible without scrolling.
- If an agent fails, show retry state without losing the completed turns.

11.6 Evidence view

- Claim ledger with status: verified, user-provided, inference, assumption, unknown or contradicted.
- Filter by agent, round, source and impact.
- Open the source excerpt that supports each claim.
- Display evidence freshness and retrieval time when external research is enabled.
- Allow user to mark a claim incorrect and provide a correction.

11.7 Full decision analysis

- Verdict and confidence.
- Strongest thesis and antithesis arguments.
- Verified facts and disputed claims.
- User constraints used in the decision.
- Conditions under which the verdict should change.
- Risks, missing information and next action.
- Exact reply or action template when supplied by the Expert Pack.
- Reopen debate with changed assumptions.

11.8 History and search

Search titles, message content, claims, verdicts and tags. Filters include Expert Pack, date, verdict, confidence and status. Opening a history item restores the exact session state and model/provider metadata.

11.9 Settings

- Profile and language preferences.
- Theme: light, dark, system.
- Model-provider credentials stored securely.
- Default rounds, tone, cost and time limits.
- Data retention and delete-account controls.
- Expert Pack list and versions.
- Voice provider settings when enabled.

12. Visual Design System and Accessibility

12.1 Approved visual direction

The default interface is a calm light theme using off-white surfaces, cool grey borders, dark navy text and restrained semantic agent colours. The layout should feel like a modern SaaS workspace with subtle IRC/channel DNA, not a literal retro terminal skin.

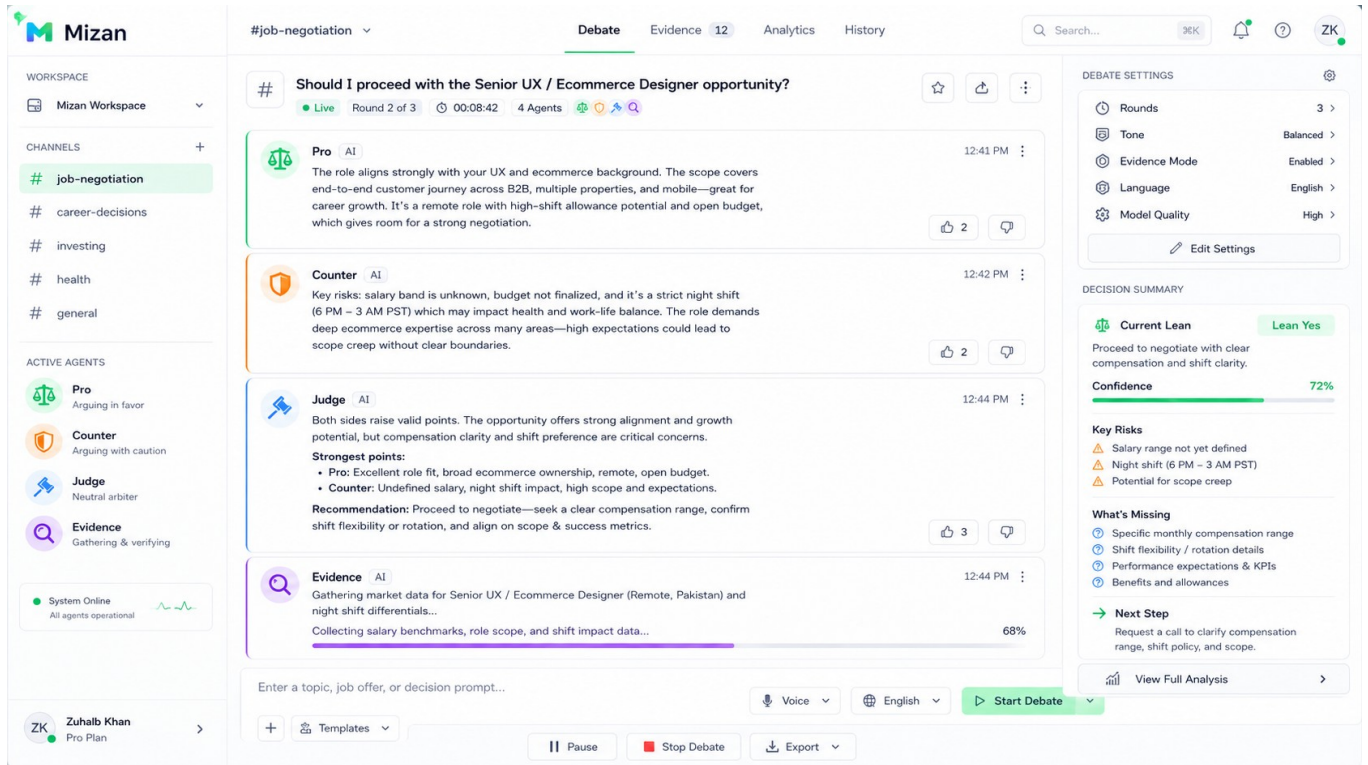


Figure: Approved light-theme direction for the primary debate channel.

12.2 Colour tokens

Token	Value	Use
--color-bg	#F7F9FC	App background
--color-surface	#FFFFFF	Cards and rails
--color-text	#132037	Primary text
--color-muted	#506075	Secondary text
--color-border	#D2DBE8	Borders and separators
--color-thesis	#14B874	Thesis / positive agent
--color-antithesis	#F48A2C	Counter / caution agent
--color-judge	#3577FF	Judge / decision
--color-evidence	#874AE6	Evidence / verification
--color-danger	#D94141	Stop, destructive and critical errors

12.3 Typography

- English interface: a clean modern sans-serif such as Inter, Geist or the project system stack.
- Urdu interface: use a swappable --font-urdu token. The approved visual direction is a clean, rounded, modern Naskh/Arabic style matching the user-provided reference. The exact family remains pending and must not be hard-coded.
- Roman Urdu uses the English sans-serif.
- Long-form messages use a comfortable line length of approximately 65-85 characters and 1.5-1.65 line-height in the browser.
- Use tabular numerals for timers, confidence and cost.

12.4 RTL support

When Urdu is selected, the entire content and navigation shell must support right-to-left direction. Agent order, icon placement, message alignment, composer and side rails must mirror logically. Do not simply right-align text inside an unchanged left-to-right layout.

12.5 Accessibility

- Target WCAG 2.2 AA contrast and keyboard behaviour.
- All controls have visible focus states and accessible names.
- Agent colours never carry meaning without labels or icons.
- Live streaming uses polite ARIA announcements without reading every token.
- Pause and stop are keyboard reachable at all times.
- Reduced-motion mode disables decorative streaming animations.
- Decision confidence is represented by text and progress, not colour alone.

13. Data Model

13.1 Primary entities

Entity	Key fields
User	id, email, passwordHash/authProvider, name, language, theme, createdAt
Workspace	id, ownerId, name, settingsJson, createdAt
Channel	id, workspaceId, slug, title, topic, status, expertPackId, language, createdAt
DebateSession	id, channelId, state, roundLimit, currentRound, tone, timeLimit, tokenLimit, costLimit, startedAt, completedAt
AgentConfig	id, sessionId, role, provider, model, systemPromptVersion, temperature, maxTokens
Message	id, sessionId, agentRole/userId, round, messageType, content, structuredJson, sequence, createdAt
Claim	id, messageId, text, status, confidence, decisionImpact
EvidenceItem	id, sessionId, sourceType, title, url/fileId, excerpt, status, freshness, metadataJson
ClaimEvidence	claimId, evidenceId, relation: supports/contradicts/context
Decision	id, sessionId, verdict, confidence, headline, outputJson,

Entity	Key fields
	createdAt
Attachment	id, channelId, storageKey, filename, mimeType, extractedText, status
ExpertPack	id, slug, version, manifestJson, active
UsageEvent	id, userId, sessionId, provider, model, tokensIn, tokensOut, latencyMs, estimatedCost, status

13.2 Persistence rules

- Messages are append-only; corrections create a new message or claim version.
- A session stores the exact Expert Pack and prompt versions used for reproducibility.
- Deleting a channel cascades to messages, claims, evidence and generated exports after confirmation.
- Provider API keys are never stored in plain text.
- Large attachments and exports live in object storage; metadata lives in PostgreSQL.

14. API and Real-Time Event Contracts

14.1 REST endpoints

```

POST  /api/auth/login
GET   /api/workspaces
POST  /api/workspaces
GET   /api/channels?workspaceId=...
POST  /api/channels
GET   /api/channels/:id
POST  /api/channels/:id/attachments
POST  /api/debates
POST  /api/debates/:id/start
POST  /api/debates/:id/pause
POST  /api/debates/:id/resume
POST  /api/debates/:id/stop
POST  /api/debates/:id/interject
POST  /api/debates/:id/verdict
GET   /api/debates/:id
GET   /api/debates/:id/evidence
GET   /api/debates/:id/export?format=pdf|md|json
GET   /api/expert-packs
GET   /api/usage/summary

```

14.2 WebSocket events

Event	Direction	Payload summary
session.state	server -> client	sessionId, state, round, elapsedMs
agent.started	server -> client	agentRole, messageId, round
agent.token	server -> client	messageId, delta
agent.completed	server -> client	messageId, structuredResponse
agent.failed	server -> client	agentRole, errorCode, retryable
evidence.updated	server -> client	claimId, evidence status

Event	Direction	Payload summary
judge.updated	server -> client	partial decision summary
user.interject	client -> server	text, attachments, intent optional
session.pause	client -> server	sessionId
session.stop	client -> server	sessionId
session.verdict	client -> server	sessionId, useCurrentEvidence=true

14.3 Idempotency and recovery

- Every mutation accepts an idempotency key.
- Agent jobs use a durable queue with retry counts and dead-letter status.
- On reconnect, the client requests events after the last received sequence number.
- Pausing prevents new agent turns but does not corrupt a provider response already received.
- Stopping cancels queued work and marks in-flight requests cancelled when provider support permits.

15. Model-Provider Abstraction

15.1 Adapter interface

```
interface ModelProviderAdapter {
  providerId: string;
  listModels(): Promise<ModelDescriptor[]>;
  stream(request: ModelRequest, signal: AbortSignal): AsyncIterable<ModelDelta>;
  completeJson<T>(request: StructuredModelRequest<T>): Promise<T>;
  estimateCost(usage: TokenUsage, modelId: string): MoneyEstimate;
  supports(feature: 'streaming' | 'json' | 'tools' | 'vision' | 'audio'): boolean;
}
```

15.2 Provider rules

- At least Thesis and Antithesis must be configurable independently. They may use different providers or different models from one provider.
- The judge should default to the highest-reliability configured model, but users may override it.
- Provider-specific request and response objects may not leak into orchestration code.
- All model responses pass through schema validation before persistence.
- If structured validation fails, request one repair turn, then mark the agent failed and expose retry.

15.3 Context assembly

Each agent receives only the context needed for its role: topic, user constraints, Expert Pack, shared evidence bundle, relevant prior arguments and round instruction. Do not resend the entire raw transcript when a structured summary and claim ledger will preserve context at lower cost.

16. Safety, Privacy and Trust

16.1 Trust labels

Label	Meaning
Verified	Supported by a visible evidence item.
User provided	Explicitly supplied by the user but not independently verified.
Inference	Reasoned conclusion from stated facts.
Assumption	Used temporarily because information is missing.
Unknown	No reliable basis available.
Contradicted	Evidence conflicts with the claim.

16.2 Privacy requirements

- TLS for all traffic and secure HTTP-only session cookies.
- Encrypt provider credentials and sensitive configuration at rest.
- No user content in logs by default; log IDs, states and error metadata.
- Clear retention controls and user-initiated deletion.
- Attachment malware scanning or strict file-type and size controls before processing.
- Prevent prompt injection inside uploaded evidence from changing system or agent roles.
- Keep system prompts and provider keys server-side only.

16.3 Prompt-injection defence

Uploaded files, URLs and user-pasted recruiter messages are untrusted evidence. The evidence extractor must wrap them as quoted source material. Agent prompts explicitly state that instructions contained inside evidence are content to analyse, not commands to execute.

16.4 Decision disclaimer behaviour

Do not show a generic disclaimer on every message. Show contextual boundaries only when risk requires them. For high-stakes topics, the final result must state what AI can and cannot determine and what qualified professional input is still needed.

17. Analytics and Observability

17.1 Product analytics

- Debates created, started, completed, stopped and reopened.
- Time from topic creation to first useful verdict.
- Clarification questions asked and completion rate.
- User intervention rate by round.
- Export and copy-exact-reply usage.
- Verdict feedback: useful, not useful, changed my decision.
- Expert Pack adoption and failure rates.

17.2 Operational metrics

- Latency to first token and total agent latency.
- Token and estimated cost by provider, model, agent and session.
- Schema validation failures and repair success.
- Retry, timeout and cancellation rates.
- WebSocket reconnects and missed event recovery.
- Queue depth and stalled jobs.

17.3 Audit trail

For each decision, retain the topic, context version, Expert Pack version, model/provider configuration, full message sequence, evidence ledger and final decision JSON. The user must be able to export this trail.

18. Testing and Acceptance Criteria

18.1 Test layers

Layer	Coverage
Unit	Router rules, scoring, state transitions, schema validation, cost calculations.
Integration	Database, queue, provider mocks, attachments, export and auth.
Contract	Every provider adapter against the internal request/response contract.
End-to-end	Create channel -> clarify -> run three rounds -> interject -> verdict -> export.
Visual regression	Light/dark themes, 1440 desktop, 1024 tablet and mobile-responsive widths.
Accessibility	Keyboard, focus, screen-reader labels, contrast and reduced motion.
Resilience	Provider timeout, malformed JSON, disconnect, pause, stop and recovery.
Safety	Prompt injection, unsupported expert claims, high-stakes routing and data deletion.

18.2 MVP acceptance criteria

27. A new user can create a Career and Hiring decision channel and start a debate without developer assistance.
28. Thesis and Antithesis stream visibly and use independently configurable models.
29. The user can pause, resume, stop and interject without losing prior messages.
30. The system asks clarification questions when required fields are missing.
31. The judge returns a valid structured decision with confidence, risks, missing information and next action.
32. Claims display trust labels and link to supplied evidence when available.
33. The final session can be reopened and exported as Markdown, JSON and PDF.
34. Light theme matches the approved information architecture and remains readable at 1440x900.

35. Urdu mode applies RTL to the full shell and uses the swappable Urdu font token.
36. Provider failure produces a recoverable error state, not a blank or corrupted channel.
37. No current salary, company claim or market figure is invented by the HR Expert Pack.
38. All core flows pass automated end-to-end tests.

19. Deployment and Operations

19.1 Recommended production stack

Layer	Recommendation
Monorepo	pnpm workspaces with a task runner such as Turborepo.
Web	Next.js with React and TypeScript.
API / realtime	TypeScript server using Fastify or NestJS plus WebSocket support.
Database	PostgreSQL with Prisma or equivalent typed ORM.
Queue / coordination	Redis and BullMQ or equivalent durable queue.
Object storage	S3-compatible storage; local filesystem adapter for development.
Styling	Tailwind CSS plus reusable accessible components; no page-specific inline styling.
Validation	Zod or equivalent shared schemas.
Testing	Vitest/Jest, Playwright and provider mocks.
Deployment	Docker Compose on the user's VPS, reverse-proxied with Caddy or Nginx.
Observability	Structured logs, error tracking and metrics endpoint.

19.2 Repository structure

```

mizan/
  apps/
    web/           # Next.js UI
    api/           # REST + WebSocket + orchestration API
    worker/        # queued agent/evidence/export jobs
  packages/
    ui/            # shared design system
    core/          # domain types, state machine, scoring
    prompts/       # versioned agent prompt templates
    model-adapters/ # provider-agnostic adapters
    expert-packs/  # career-hiring and future packs
    schemas/       # shared Zod/JSON schemas
    config/        # environment and feature flags
  infrastructure/
    docker/
    migrations/
    scripts/
  tests/
    e2e/
    fixtures/
  docs/
    architecture/
    decisions/

```

19.3 Environments

- Local: Docker Compose with mock provider mode and seeded HR debate.
- Staging: separate database and provider keys; anonymised fixtures only.
- Production: HTTPS, backups, rate limits, encrypted secrets and monitored queue workers.
- Feature flags for external research, voice, team sharing and additional Expert Packs.

19.4 Backups and failure recovery

- Daily PostgreSQL backup with retention policy.
- Object-storage lifecycle and export cleanup.
- Redis is not the source of truth; queued jobs can be reconstructed from database state.
- Debate session states are resumable after process restart.
- Health endpoints for web, API, worker, database and queue.

20. Milestones and Codex Delivery Plan

20.1 Implementation order

Phase	Deliverables	Exit condition
0 - Foundation	Monorepo, Docker, database, auth, shared schemas, CI.	Local stack starts with one command and tests pass.
1 - Product shell	Light UI, workspace, channels, history, create-decision modal.	Approved responsive shell with seeded data.
2 - Runtime	State machine, queue, WebSocket streaming, pause/stop/interject.	Two mock agents complete a bounded debate.
3 - Models	Two provider adapters, structured validation, token/cost tracking.	Real models can be configured without orchestration changes.
4 - Expert system	Router, clarifier, Expert Pack loader, Career and Hiring pack.	Recruiter-message scenario completes correctly.
5 - Evidence and judge	Attachments, claim ledger, evidence labels, verdict contract.	Final decision links arguments and evidence.
6 - Export and quality	PDF/MD/JSON export, RTL, dark theme, accessibility, E2E.	All MVP acceptance criteria pass.
7 - Deploy	Production Docker Compose, reverse proxy, backups, runbook.	Application available securely on VPS.

20.2 Codex working rules

- Implement vertical slices that remain runnable after every phase.
- Do not replace structured agent contracts with raw text-only prompts.
- Do not hard-code model names, provider request formats or HR rules inside UI components.
- Do not invent additional screens or features before completing mandatory acceptance criteria.
- Write migrations, seeds, tests and README instructions alongside the feature that requires them.
- Use mock provider adapters so the full application is testable without paid API calls.
- Keep generated text safe to render; use Markdown sanitisation and never inject raw HTML from models.

- Ask for a product decision only when this document truly leaves an implementation blocker.

20.3 Definition of done

MVP is done when

A user can sign in, create a career decision, provide recruiter context, receive clarification, watch two different agents debate in real time, inspect evidence labels, obtain a judge verdict, intervene or stop at any time, reopen the decision later and export the complete record. The product must run from documented Docker commands on the target VPS.

Appendix A - Seed Debate for Development

Seed the application with the following channel so Codex can verify the full interface and runtime:

```
Channel: #job-negotiation
Topic: Should I proceed with the Senior UX / Ecommerce Designer opportunity?
Context:
- Recruiter contacted the user directly.
- Remote role at ATC Holdings.
- Pakistan hours: 6 PM to 3 AM.
- Scope includes B2B ecommerce journeys, product pages, checkout, delivery communication, mobile UX and consistency across multiple properties.
- Recruiter says the budget is open and shortlisted candidates may receive more than current earnings.
- The user does not want current salary used as the offer benchmark.
Desired output:
- Whether to take the first call.
- What not to disclose.
- Exact next message.
- Questions to ask during screening.
```

Appendix B - Sample Final Decision

```
Verdict: PROCEED CONDITIONALLY
Confidence: 0.78
Reason: The role is sufficiently relevant to justify a low-cost screening call, but permanent night hours and undefined compensation require clarification.
Conditions:
1. Do not disclose current salary as the compensation benchmark.
2. Confirm direct employment versus agency contract.
3. Ask for total package, benefits and performance expectations.
4. Stop the process if the final package does not compensate for the fixed night schedule.
Next action: Schedule a 15-minute screening call and use the prepared salary-boundary response.
```

Appendix C - UI Reference Notes

- The approved primary direction is the light theme shown in this document.
- The dark theme remains an optional user preference, not the default.
- The product must preserve the three-column structure at wide desktop sizes but collapse the right rail into a drawer at narrower widths.
- The central feed is the primary content and must receive the largest width.

- Agent cards are visually distinct but should not become large decorative tiles that reduce reading density.

The screenshot displays the Mizan Decision Room interface. The main area shows a debate titled "#job-negotiation" with the question "Should I proceed with the Senior UX / Ecommerce Designer opportunity?". The debate is in progress, with 4 agents participating. The interface includes a sidebar with workspace and channels, active agents (Pro, Counter, Judge, Evidence), and a decision summary on the right. The decision summary shows a "Current Lean" towards "Lean Yes" with a confidence of 72%. Key risks include salary range not yet defined, night shift (6 PM - 3 AM PST), and potential for scope creep. The next step is to request a call to clarify compensation range, shift policy, and scope.

Optional dark-theme reference.

Appendix D - Final Handoff Checklist

Item	Required artifact
Source code	Complete repository with clean commit history or release archive.
Environment	Example environment file with every required variable documented.
Database	Migrations and seed data.
Providers	Mock adapter plus setup guide for configured real providers.
Deployment	Docker Compose, reverse-proxy config and VPS runbook.
Quality	Unit, integration and Playwright tests with commands.
UX	Responsive light/dark UI, RTL Urdu support and accessibility checks.
Documentation	README, architecture note, Expert Pack authoring guide and troubleshooting.
Demo	Seeded #job-negotiation debate completing end-to-end.

Final product boundary

Mizan is not a two-chat copy machine. It is a controlled decision runtime. Its value comes from routing, expert packs, evidence discipline, bounded disagreement, calibrated judgment and a practical action record. Preserve that distinction throughout implementation.